Hints: Functional and differential equations – 1. Functional equations: the toolkit

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. On $\mathbb{Q}$ additivity gives $f(x) = cx$, and $f(1) = f(1)^2$ gives $c \in \{0, 1\}$.

Hint 2. On $\mathbb{R}$: $f(x^2) = f(x)^2 \geqslant 0$, so $f \geqslant 0$ on $[0, \infty)$, and additivity makes f non-decreasing. A monotone additive function is linear. Answer in both cases: $f \equiv 0$ and $f(x) = x$.

Problem 2

Hint 1. At $x = 0$ the inequality is an equality: $f(y) \geqslant f(y) + 2^y f(0)$. So for fixed y the function $x \mapsto f(x + y) - 3^x f(y) - 2^y f(x)$ has a minimum 0 at $x = 0$, and its derivative vanishes there.

Hint 2. This gives $f'(y) = f(y) \ln 3 + 2^y f'(0)$. In the same way (minimum in y at $y = 0$) $f'(x) = 3^x f'(0) + f(x) \ln 2$.

Hint 3. Subtract: $f(x) \ln \frac{3}{2} = f'(0)(3^x - 2^x)$, and $f(1) = 1$ fixes $f'(0) = \ln \frac{3}{2}$. Answer: $f(x) = 3^x - 2^x$, which satisfies the inequality with equality.

Problem 3

Hint 1. Let $g(x) = x^2 - 2$. If $f \circ f = g$, then $f \circ g = f \circ f \circ f = g \circ f$: f maps the fixed points of g to fixed points of g, and the fixed points of $g \circ g$ to fixed points of $g \circ g$.

Hint 2. g has the fixed points $2, -1$, and $g \circ g$ has in addition the two points $\frac{-1 \pm \sqrt{5}}{2}$, which g swaps. Where can f send $b = \frac{-1+\sqrt{5}}{2}$? If $f(b) \in \{2, -1\}$, then $g(b) = f(f(b)) \in \{2, -1\}$; if $f(b) = b$, then $g(b) = b$; if $f(b) = b'$, then $f(b') = g(b) = b'$ and $g(b') = f(f(b')) = b'$. Each case contradicts $g(b) = b'$, $g(b') = b$, so no f exists.

Hint 3. For $x^2 + 1$ such f exist. On $[0, \infty)$ the map $g(x) = x^2 + 1$ is increasing without fixed points, and $0, \frac{1}{2}, 1, g(\frac{1}{2}), g(1), \ldots$ cut $[0, \infty)$ into consecutive intervals. Take any increasing bijection $h\colon [0, \frac{1}{2}) \to [\frac{1}{2}, 1)$, put $f = h$ on $[0, \frac{1}{2})$, $f = g \circ h^{-1}$ on $[\frac{1}{2}, 1)$, and extend by $f(g(x)) = g(f(x))$. Finally $f(x) = f(|x|)$ for $x < 0$.

Problem 4

Hint 1. Pair the factors: $\int f^k \cdot \int f^{2021-k}$, $k = 1, \ldots, 1010$. Compare each pair with $\int f^{2021}$.

Hint 2. Chebyshev's inequality for similarly ordered functions: $\int_0^1 f^a \int_0^1 f^b \leqslant \int_0^1 f^{a+b}$, since $\iint (f^a(x) - f^a(y))(f^b(x) - f^b(y))\, dx\, dy \geqslant 0$. Equality holds only if the integrand vanishes, i.e. f is constant.

Hint 3. So the left-hand side is at most the right-hand side, with equality only for constants. Answer: $f \equiv c$, $c \geqslant 0$.

Problem 5

Hint 1. $f(3n) = f(f(f(n))) = 3f(n)$. Also $f(1) \neq 1$ and $f(1) = b$ gives $f(b) = 3 > b$, so $f(1) = 2$, $f(2) = 3$, and by induction $f(3^k) = 2 \cdot 3^k$, $f(2 \cdot 3^k) = 3^{k+1}$.

Hint 2. Between 3^k and $2\cdot 3^k$ there are as many integers as between $2\cdot 3^k$ and 3^{k+1}, and f is strictly increasing, so $f(3^k{+}j) = 2{\cdot}3^k{+}j$ for $0 \leqslant j \leqslant 3^k$; then $f(2{\cdot}3^k{+}j) = f(f(3^k{+}j)) = 3^{k+1}{+}3j$. This proves uniqueness.

Hint 3. For $f(f(n)) = 2n$: $f(1) = b$ gives $f(b) = 2$. A strictly increasing $f\colon \mathbb{N} \to \mathbb{N}$ has $f(n) \geqslant n$, so $b \leqslant 2$; $b = 1$ gives $f(f(1)) = 1$, and $b = 2$ gives $f(2) = 2 = f(1)$. So there is no strictly increasing solution (without monotonicity there are many).

Problem 6

Hint 1. An additive f is $\mathbb{Q}$-linear, so its graph Γ is a $\mathbb{Q}$-subspace of $\mathbb{R}^2$. If f is not linear, there are x, y with $\frac{f(x)}{x} \neq \frac{f(y)}{y}$, so Γ contains two $\mathbb{R}$-linearly independent vectors.

Hint 2. The $\mathbb{Q}$-span of two $\mathbb{R}$-independent vectors of $\mathbb{R}^2$ is dense. Hence a non-dense graph means f is linear.

Hint 3. If f is measurable, some set $E = \{|f| \leqslant M\}$ has positive measure, and by Steinhaus $E - E$ contains an interval $(-\delta, \delta)$. There $|f| \leqslant 2M$, so the graph misses a whole box, and f is linear.

Problem 7

Hint 1. Put $a = f(0)$, $b = f(\frac{\pi}{2})$ and use three substitutions: $(x, y) = (0, t)$, $(\frac{\pi}{2}+t, \frac{\pi}{2})$, $(\frac{\pi}{2}, \frac{\pi}{2}+t)$.

Hint 2. They give $f(t)+f(-t) = 2a\cos t$, $f(\pi+t)+f(t) = 0$, $f(\pi+t)+f(-t) = -2b\sin t$. Add the first two and subtract the third: $f(t) = a\cos t + b\sin t$. These work. No regularity is needed because the answer comes from finitely many substitutions, without any limit.

Problem 8

Hint 1. First equation: $x = y = 0$ gives $f(0) \in \{0, 1\}$, and $y = 0$ gives $f(0)(f(x) - 1) = x$. So $f(0) = 1$ and $f(x) = x + 1$, which works.

Hint 2. Second equation: $f(0) = f(1) = f(-1) = 0$ and $f(-x) = -f(x)$. For $x > 0$ put $h(t) = \frac{f(e^t)}{e^t}$: then $h(s+t) = h(s) + h(t)$.

Hint 3. Answer: $f(x) = x\,A(\ln|x|)$ for $x \neq 0$ and $f(0) = 0$, where A is any additive function; the regular ones are $f(x) = cx\ln|x|$.

Problem 9

Hint 1. Some regularity is necessary: a non-linear additive function is subadditive, but $\frac{f(x)}{x}$ has no limit. Let $L = \inf_{x>0} \frac{f(x)}{x} \in [-\infty, \infty)$; clearly $\liminf \frac{f(x)}{x} \geqslant L$.

Hint 2. Fix $a > 0$ and write $x = na{+}r$ with $r \in [a, 2a)$. Subadditivity gives $f(x) \leqslant nf(a){+}f(r) \leqslant nf(a) + M_a$, where M_a is an upper bound of f on $[a, 2a]$. Dividing by x and letting $x \to \infty$: $\limsup \frac{f(x)}{x} \leqslant \frac{f(a)}{a}$ for every $a > 0$, i.e. $\limsup \leqslant L$.

Hint 3. This is the continuous version of Fekete's lemma: the sequence $a_n = f(n)$ is subadditive, and the local upper bound replaces the integer step.

Problem 10

Hint 1. Put $x = t + h$, $y = t - h$: $f(t+h) - f(t-h) = 2hg(t)$. Integrating over $h \in [0, 1]$ gives $g(t) = \int_t^{t+1} f - \int_{t-1}^{t} f$, so $g \in C^1$; and $f(x) = f(0) + xg(\frac{x}{2})$ gives $f \in C^1$. Repeating, $f, g \in C^\infty$.

Hint 2. Now differentiate $f(t+h) - f(t-h) = 2hg(t)$ in h: once, at $h = 0$, gives $g = f'$; twice gives $f''(t+h) = f''(t-h)$ for all t, h, so f'' is constant.

Hint 3. Answer: $f(x) = ax^2 + bx + c$, $g(x) = 2ax + b$.